

items on time. With the help of a good SCM system, one can get the transit information of a shipment and track it from point to point. Artificial intelligence (AI) can also help to control the SCM to get better results and it requires less human effort. However, any SCM requires storage of data in a system like a centralised server or cloud storage, but such systems could be vulnerable to virus or hacking attacks. A fraudulent interception of any shipment can happen on a centralised SCM system. As blockchain is always decentralised, no such fraudulent activity can happen easily on a SCM that uses a blockchain network. During a pandemic like COVID-19, there is a need for essential goods around the globe and this demand can only be fulfilled by reliable logistics, backed by a SCM system that uses blockchain. Many logistics companies like Maersk, FedEx, DHL and UPS are using blockchain for their operations.

- **Infection tracking:** During a pandemic, tracking individuals who are infected is very important and this is done with the help of various screening processes. For COVID-19, the screening for infection is done at various entry points like airports and railway stations using thermal scanners. Then the infected person is sent into quarantine. These quarantined people are digitally marked through their smartphones. That's why some mobile apps can help to identify people who are already quarantined or diagnosed as having COVID-19.

Apps like Aarogya Setu can help people maintain social distance and detect any infected person within close range. Such an app uses Bluetooth and GPS to detect any infected person who is also using the same app. Some apps can even tell which hospitals will admit COVID-19 patients and how many beds are available. However, such apps use a centralised or a cloud based network but are not always reliable. A blockchain network that

stores data about COVID-19 patients is safe and reliable. As every block in a blockchain is tamper-proof, the data stored on such a network is always safe and secure. As blockchain networks are less vulnerable to fraudulent attacks, the infection tracking information stored on such networks is not affected in any manner. Moreover, most infection tracking systems used by the government are centralised and cannot be used outside the country. Such systems have limitations and if an infected person moves outside the country, then this move will not be detected by other countries. As blockchain is a decentralised system, it can be used by people around the globe and the information about an infected individual can be shared with every user of the blockchain. This would enable authorities to detect a COVID-19 infected person, even if that person travels to another country. Therefore, using blockchain for infection tracking can be very useful in pandemics like COVID-19.

Pros and cons of blockchain

Pros

Decentralised: The decentralised network of a blockchain allows the users to control it, instead of a centralised authority. This prevents any unfair activity over the network and gives equal rights to all its users.

Transparency: Blockchain is a transparent network, and all the stored data is shared with every user or node in the network. This makes the data transparent to every user and prevents any data tampering. It is especially useful for cryptocurrencies and if any fraudulent transaction is done by an attacker, then everyone in the network will know about it.

Anonymity: Blockchain uses a network in which every user remains anonymous and this

gives full privacy to the users. Such anonymity also protects the users from any kind of identity theft. Hence, all types of users including big organisations can use a cryptocurrency of a blockchain to do any kind of transaction.

Security: As blockchain uses a decentralised network and every user is anonymous, it is difficult for hackers or computer viruses to attack the network easily. Moreover, cryptocurrencies that use proof-of-work (PoW) for the blockchain are more secure and transactions are more reliable.

Cons

Performance: The decentralised nature of a blockchain makes it a slow and time consuming network. When compared to other cloud based networks, the speed and performance of any blockchain network can vary depending on the users and the computers used by them. In the case of cryptocurrencies like Bitcoin, the speed of the miners can affect the performance of the network.

Power consumption: The power consumption of the ASIC machines used for mining can vary depending on the processing power. The higher the processing power of the machine, the more power it will consume.

Blockchain is a unique way of storing data over a network, and because of its exceptional features, it's been accepted by many companies for their products and services. Even though blockchain has a few drawbacks, its benefits and advantages outweigh its disadvantages. Therefore, blockchain can be very useful during a pandemic like COVID-19. 

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